山东大学 计算机科学与技术 学院

计算机网络 课程实验报告

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实验题目：计算机网络实验 6		
实验学时：2	实验日期：2024.4.24	
实验目的：详细研究了解 NAT 协议		

实验结果：

1、 客户端 IP 地址为：192.168.1.100

o.	Time	Source	Destination	Pro
7	04:43:01.477175	192.168.1.100	74.125.91.113	HTT
11	04:43:01.543197	74.125.91.113	192.168.1.100	HTT

2、 输入后的结果如下图

http&&ip.addr == 64.233.169.104

No.	Time	Source	Destination	Protocol	Lengtl	Info
56	04:43:07.378402	192.168.1.100	64.233.169.104	HTTP	689	GET / HTTP/1.1
60	04:43:07.427932	64.233.169.104	192.168.1.100	HTTP	814	HTTP/1.1 200 OK (text/html)
62	04:43:07.550534	192.168.1.100	64.233.169.104	HTTP	719	GET /intl/en_ALL/images/logo.gif HTTP/1.1
73	04:43:07.618586	64.233.169.104	192.168.1.100	HTTP	226	HTTP/1.1 200 OK (GIF89a)
75	04:43:07.639320	192.168.1.100	64.233.169.104	HTTP	809	GET /extern_js/f/CgJlbhICdXMrMAo4NUAILCswDjgHLCswFjg
92	04:43:07.717784	64.233.169.104	192.168.1.100	HTTP	648	HTTP/1.1 200 OK (text/javascript)
94	04:43:07.761459	192.168.1.100	64.233.169.104	HTTP	695	GET /extern_chrome/ee36edbd3c16a1c5.js HTTP/1.1
100	04:43:07.806488	64.233.169.104	192.168.1.100	HTTP	870	HTTP/1.1 200 OK (text/html)
107	04:43:07.921971	192.168.1.100	64.233.169.104	HTTP	712	GET /images/nav_logo7.png HTTP/1.1
112	04:43:07.951496	192.168.1.100	64.233.169.104	HTTP	806	GET /csi?v=3&s=webhp&action=&tran=undefined&e=17259,
119	04:43:07.954921	64.233.169.104	192.168.1.100	HTTP	1359	HTTP/1.1 200 OK (PNG)
122	04:43:07.978625	192.168.1.100	64.233.169.104	HTTP	670	GET /favicon.ico HTTP/1.1
124	04:43:08.006918	64.233.169.104	192.168.1.100	HTTP	269	HTTP/1.1 204 No Content
127	04:43:08.032636	64.233.169.104	192.168.1.100	HTTP	1204	HTTP/1.1 200 OK (image/x-icon)

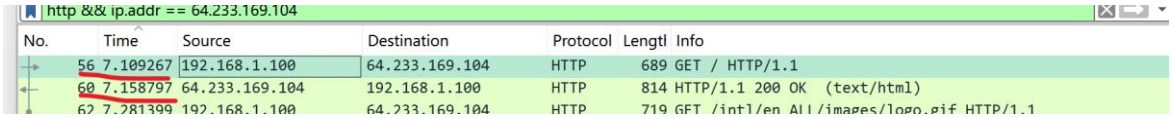
3、源地址 192.168.1.100 目标地址 64.233.169.104 源端口 4335 目标端口 80

```
> Ethernet II, Src: HonHaiPrecis_0d:ca:8f (00:22:68:0d:ca:8f), Dst: CiscoLinks
v Internet Protocol Version 4, Src: 192.168.1.100, Dst: 64.233.169.104
    0100 .... = Version: 4

v Transmission Control Protocol, Src Port: 4335, Ds
    Source Port: 4335
    Destination Port: 80
    [Stream index: 2]
```

4、在 7.158797 时接收到 200 OK 消息。源 IP 地址 64.233.169.104 目

标 IP 地址 192.168.1.100 源端口号 80 目标端口号 4335




```
[Header checksum status: Unverified]
Source Address: 64.233.169.104
Destination Address: 192.168.1.100

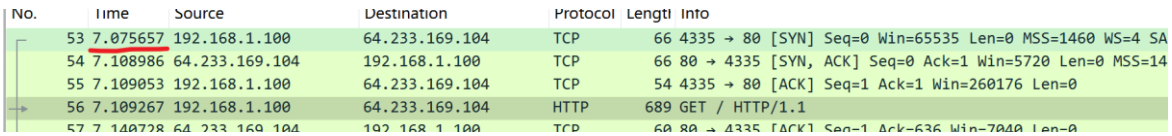
Transmission Control Protocol, Src
Source Port: 80
Destination Port: 4335
```

5、TCP SYN 段在 7.075657 时发送

TCP SYN 段 源 IP 地址为 192.168.1.100, 目标 IP 地址为 64.233.169.104 源端口号为 4335 目标端口号为 80

响应 SYN 发送的确认字符的 源 IP 地址为 64.233.169.104 目标 IP 地址为 192.168.1.100 源端口号为 80 目标端口号为 4335

Arrival Time: Sep 21, 2009 04:43:07.378121000 中国标准时间




```
Protocol: TCP (6)
Header Checksum: 0xabbb [validation disabled]
[Header checksum status: Unverified]
Source Address: 192.168.1.100
Destination Address: 64.233.169.104

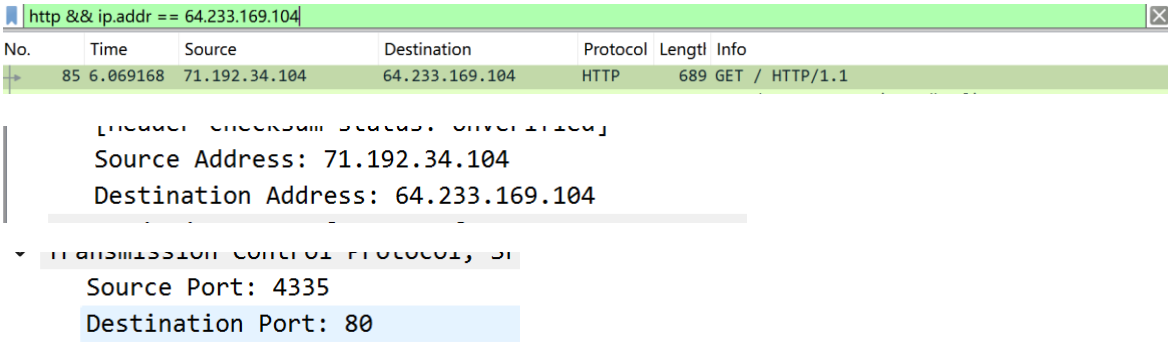
Transmission Control Protocol, Src Port: 4335, Dst Port: 80, Seq: 0, Len: 0
Source Port: 4335
Destination Port: 80
[Stream index: 2]
[Conversation completeness: Incomplete, DATA (15)]
Header Checksum: 0xe62b [validation disabl
[Header checksum status: Unverified]
Source Address: 64.233.169.104
Destination Address: 192.168.1.100
```

```
Transmission Control Protocol, Src Port: 80, Dst Port: 4335, Seq: 0, Ack:
    Source Port: 80
    Destination Port: 4335
    [Stream index: 2]

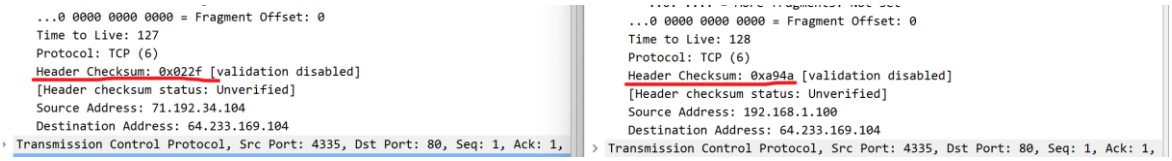
Frame 54: 66 bytes on wire (528 bits), 66 bytes captured (528 bits)
    Encapsulation type: Ethernet (1)
    Arrival Time: Sep 21, 2009 04:43:07.378121000 中国标准时间
    UTC Arrival Time: Sep 20, 2009 20:43:07.378121000 UTC
    Epoch Arrival Time: 1253479387.378121000
    [Time shift for this packet: 0.000000000 seconds]
```

6、在 6.069168 时出现源地址是 71.192.34.104 目标地址是 64.233.169.104 源端口号 4335 目标端口号 80。

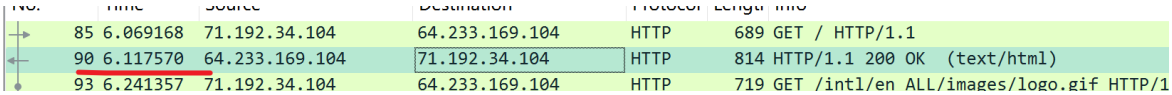
源地址不同， 目标地址、源端口号、目标端口号相同。



7、经过对比发现 Checksum 发生了变化。因为源 IP 地址发生了变化，所以校验和要重新计算，所以发生了改变。



8、6.117570 时接收到第一个 200 OK 消息，源 IP 地址为 64.233.169.104 目标 IP 地址为 71.192.34.104，源端口号为 80，目标端口号为 4335。与问题 4 对比，目标 IP 地址发生了变化，其余没有改变。



```
    [Header checksum status: Unverified]
    Source Address: 64.233.169.104
    Destination Address: 71.192.34.104
Transmission Control Protocol, Src Port: 80, Dst Port: 4335
    Source Port: 80
    Destination Port: 4335
```

9、客户端到服务器的 TCP SYN 段 在 6.099657 时捕获

服务器到客户端 TCP 确认字符段在 6.117078 时被捕获

客户端到服务器的 TCP SYN 段的源 IP 地址为 71.192.34.104 目标 IP 地址为 64.233.169.104 源端口号为 4335 目标端口号为 80

服务器到客户端 TCP 确认字符段的源 IP 地址为 64.233.169.104 目标 IP 地址为 71.192.34.104 源端口号为 80，目标端口号为 4335

与问题 5 相比 TCPSYN 段 源 IP 地址不同， 响应 SYN 发送的确认字符段目标 IP 地址不同，其余相同。

```
87 6.099637  64.233.169.104   71.192.34.104   TCP    60 80 → 4335 [ACK] Seq=1 Ack=636 Win=7040 Len=0
88 6.117078  64.233.169.104   71.192.34.104   TCP    1484 80 → 4335 [ACK] Seq=1 Ack=636 Win=7040 Len=1430 [T…
```

```
    [Header checksum status: Unverified]
    Source Address: 71.192.34.104
    Destination Address: 64.233.169.104
Transmission Control Protocol, Src Port: 433
    Source Port: 4335
    Destination Port: 80

    [Header checksum status: Unverified]
    Source Address: 64.233.169.104
    Destination Address: 71.192.34.104
Transmission Control Protocol, Src Port: 80,
    Source Port: 80
    Destination Port: 4335
    [Stream index: 21]
```

10 NAT translation table

WAN	LAN
IP addr 64.233.169.104 Port 80	IP addr 192.168.1.100 Port 4338

问题及收获：

之前曾经见到过公网，私网等的概念，在本次实验中更深的认识到了公网和私网。私网想要与公网对接，需要使用一个在公网中唯一的 IP 地址。那么私网中的其他 IP 地址就被 NAT 抽象为了多进程中的端口。使用公网唯一 IP+端口的这样一个形式完成了私网接入公网。